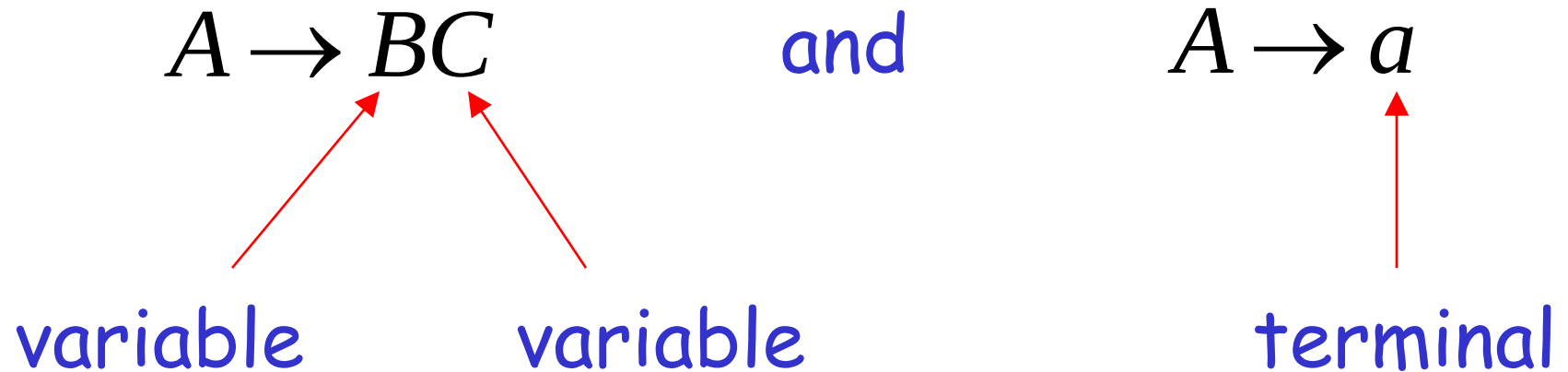


Normal Forms for Context-free Grammars

Chomsky Normal Form

All productions have form:



Examples:

$$S \rightarrow AS$$

$$S \rightarrow a$$

$$A \rightarrow SA$$

$$A \rightarrow b$$

Chomsky
Normal Form

$$S \rightarrow AS$$

$$S \rightarrow AAS$$

$$A \rightarrow SA$$

$$A \rightarrow aa$$

Not Chomsky
Normal Form

Conversion to Chomsky Normal Form

Removing All

Step 1: Remove Nullable Variables

Step 2: Remove Unit-Productions

Step 3: Remove Useless Variables

Conversion to Chomsky Normal Form

Example: $S \rightarrow ABa$

$A \rightarrow aab$

$B \rightarrow Ac$

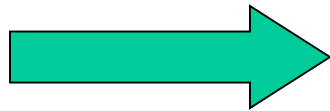
Not Chomsky
Normal Form

Introduce variables for terminals: T_a, T_b, T_c

$$S \rightarrow ABa$$

$$A \rightarrow aab$$

$$B \rightarrow Ac$$



$$S \rightarrow ABT_a$$

$$A \rightarrow T_aT_aT_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$

Introduce intermediate variable: V_1

$$S \rightarrow ABT_a$$

$$A \rightarrow T_a T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$



$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_a T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$

Introduce intermediate variable: V_2

$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_a T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$



$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_a V_2$$

$$V_2 \rightarrow T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$

Final grammar in Chomsky Normal Form:

$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_a V_2$$

$$V_2 \rightarrow T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$

Initial grammar

$$S \rightarrow ABa$$

$$A \rightarrow aab$$

$$B \rightarrow Ac$$

In general:

From any context-free grammar
not in Chomsky Normal Form

we can obtain:

An equivalent grammar
in Chomsky Normal Form

The Procedure

First remove:

Nullable variables

Unit productions

Useless Productions

For every symbol a :

Add production $T_a \rightarrow a$

In productions: replace a with T_a

New variable: T_a

Replace any production $A \rightarrow C_1 C_2 \cdots C_n$

with $A \rightarrow C_1 V_1$

$V_1 \rightarrow C_2 V_2$

$\dots$

$V_{n-2} \rightarrow C_{n-1} C_n$

New intermediate variables: $V_1, V_2, \dots, V_{n-2}$

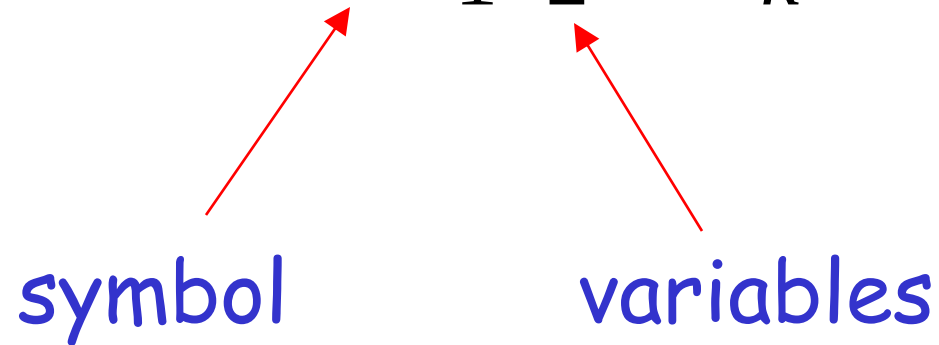
Theorem: For any context-free grammar there is an equivalent grammar in Chomsky Normal Form

Observations

- Chomsky normal forms are good for parsing and proving theorems
- It is very easy to find the Chomsky normal form of any context-free grammar

Greibach Normal Form

All productions have form:

$$A \rightarrow a V_1 V_2 \cdots V_k \quad k \geq 0$$


symbol

variables

Examples:

$$S \rightarrow cAB$$

$$A \rightarrow aA \mid bB \mid b$$

$$B \rightarrow b$$

Greinbach
Normal Form

$$S \rightarrow abSb$$

$$S \rightarrow aa$$

Not Greinbach
Normal Form

Conversion to Greinbach Normal Form:

$$S \rightarrow abSb$$

$$S \rightarrow aa$$



$$S \rightarrow aT_bST_b$$

$$S \rightarrow aT_a$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

Greinbach
Normal Form

Theorem: For any context-free grammar there is an equivalent grammar in Greinbach Normal Form

Observations

- Greinbach normal forms are very good for parsing
- It is hard to find the Greinbach normal form of any context-free grammar

An Application of Chomsky Normal Forms

The CYK Membership Algorithm

Input:

- Grammar G in Chomsky Normal Form
- String w

Output:

find if $w \in L(G)$

The Algorithm

Input example:

- Grammar G : $S \rightarrow AB$

$$A \rightarrow BB$$
$$A \rightarrow a$$
$$B \rightarrow AB$$
$$B \rightarrow b$$

- String w : $aabbbb$